If two genes look interchangeable, are they? A calibration of expression redundancy against interventional equivalence across 43 million gene pairs

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Abstract

A routine step in computational biology treats two genes that track each other closely in expression data as one hypothesis: pick either, and the other would have behaved the same. We could find no published estimate of how often that holds, so we measured it. Using DepMap 24Q4—1,103 cancer cell lines with paired expression profiles and genome-wide CRISPR knockout screens—we scored 43.9 million gene pairs on two axes: how redundant the two genes look observationally, and how similar the consequences of knocking each one out actually are. Redundancy is real evidence and it is far from identification. Relative to a base rate of 0.42% (95% CI 0.37–0.47), the odds of interventional equivalence rise about 39-fold (95% CI 23–65) in the most redundant well-populated bins, but the ceiling there is 17.0% (95% CI 9.7–27.3); among the 13 most redundant pairs in the dataset, none were equivalent. Three secondary results follow. Cell lineage manufactures apparent redundancy: correcting for it removes 56% of pairs at $r^2 \geq 0.6$. Panel redundancy—predictability from ten partners rather than one—carries essentially the same information as best-pair redundancy (AUC 0.614 versus 0.607). And genes inside a gene family show a *lower* equivalence ceiling (13.9%) than genes outside one (26.3%), the sign that paralog buffering predicts. We report two pre-registered control failures in full: a null that could not fail, and a positive control whose failure correctly indicted our first similarity measure rather than the biology. Code, pre-registration and the complete experiment ledger are public.

Author summary

When researchers look for genes worth targeting in cancer, they often start from expression data: measurements of how strongly each gene is switched on across many samples. Two genes sometimes rise and fall together so closely that they look interchangeable, and it is natural to treat them as one possibility—to follow one up and assume the other would have behaved the same. As far as we could find, that assumption had never been measured.

We measured it. Using a public resource pairing expression profiles with the results of switching off each gene in turn across more than a thousand cancer cell lines, we asked: when two genes look interchangeable, how often does switching off one do what switching off the other does?

Looking interchangeable is real evidence, but weak. It raises the odds roughly forty-fold, yet even among the most redundant-looking pairs only about one in six behave the same way at best, and among the most extreme pairs in the dataset none did. Apparent redundancy is also substantially an artefact of tissue type. Researchers choosing which gene to pursue should treat expression similarity as grounds for suspicion, never for confidence.

Introduction

Given expression profiles across many samples, two genes sometimes track each other so closely that they appear interchangeable. A common next step is to treat them as a single hypothesis—to perturb one, target one, or follow one up, on the understanding that the other would have behaved similarly. The step is rarely stated explicitly, which is part of why it is rarely checked.

Stated explicitly it is a conditional probability. Let r_{obs} measure how redundant two genes look observationally, and let a pair be *interventionally equivalent* when knocking out either produces the same phenotype. The quantity the everyday decision relies on is

$$P(\text{equivalent under intervention} \mid r_{\mathrm{obs}} = r)\,,$$

as a function of r. This paper measures that function.

Adjacent literature is substantial and should not be mistaken for an answer to this question. Co-essentiality analyses ask the *reverse* direction: given that two genes have similar knockout profiles, what else do they share? That is informative about pathway membership, but says nothing about how often observational similarity licenses an interventional claim, which is the direction in which decisions are actually made.

Closer still is the work of **(author?)** [1] on paralog buffering, and its extension to genome-scale synthetic-lethality prediction [2, 3]. Those studies ask whether losing one member of a paralog pair *increases dependency* on the other. That is a question about compensation—about pairs whose members cover for one another—and it is measured on paralogs specifically. Interventional equivalence is a different relation: it asks whether the two knockouts produce the *same* outcome, over the whole gene universe rather than paralogs alone. The two are not merely different; they point opposite ways. A perfectly buffering pair is maximally non-equivalent under single knockout, precisely because each gene covers for the other. We recover that predicted opposition in our own data and report it as convergence with their finding rather than as an independent one.

We could not find the forward calibration published. We state that as the result of a search rather than as a claim about the literature, and would welcome a pointer to prior work.

Results

Controls, including the two that failed

Because the headline of this study is a number rather than a mechanism, the controls determine whether the number is worth anything. Two of the three pre-registered controls failed, and both failures were informative.

A null that could not fail. The pre-registered null permuted the rows (cell lines) of one matrix. Its output was bin-for-bin identical to the real analysis—which is how the error surfaced. Within-matrix correlations are invariant under a shared row permutation, so the control was algebraically incapable of differing from the thing it was meant to test. It was replaced with a gene-label permutation, which can fail. Under the replacement the calibration curve flattens: base rate 0.422% and a maximum bin equivalence of 0.9%, against 17% in the real data.

A positive control that failed, correctly. 253 same-complex gene pairs drawn from CORUM are the most functionally equivalent pairs biology offers. Under our first declared measure, the Pearson correlation of Chronos profiles (e_{sel}), they scored a median of 0.221 with *none* above the equivalence threshold; PSMA1–PSMB5, two subunits of the same proteasome, scored 0.04. The pre-registration directed that a failing positive control indicts the measure before the biology. It did: these genes are near-uniformly essential, so their Chronos profiles are flat and highly similar in exactly the mean that Pearson discards. Under the replacement measure e_{prox} (Methods) the same pairs score a median of 0.925 with 88.1% above threshold, and PSMA1–PSMB5 scores 0.90.

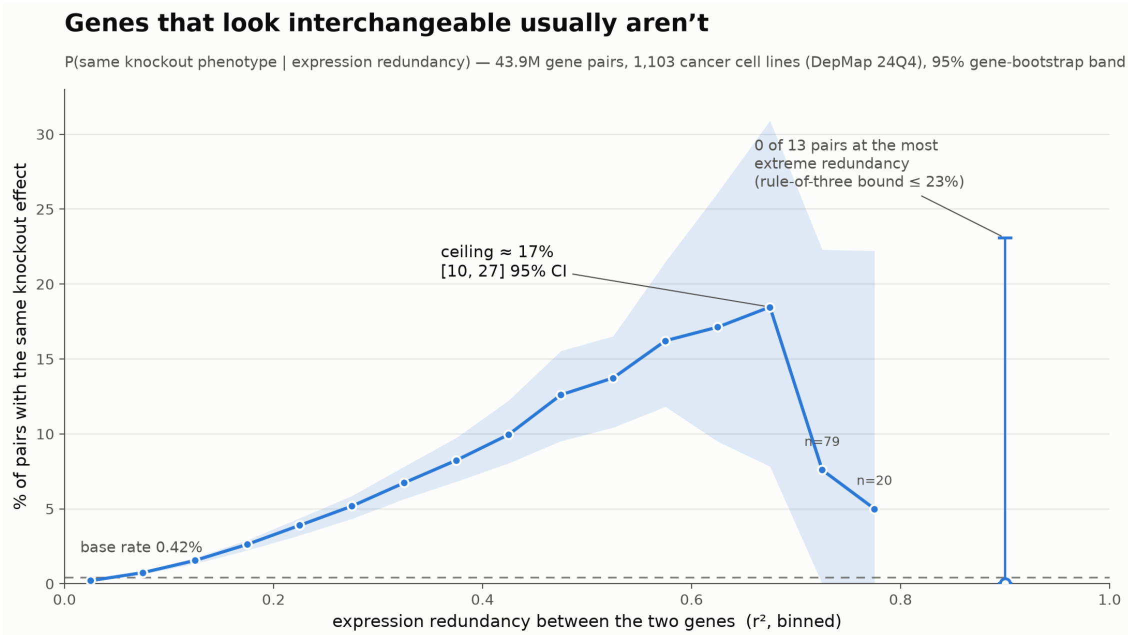

Figure 1: **Probability that two genes are interventionally equivalent, as a function of how redundant they look in expression.** Band is the 95% gene-level bootstrap interval; the dashed line is the base rate over all 43.9M pairs. The isolated point pools $r^2 \geq 0.8$, where 0 of 13 pairs are equivalent; its whisker is the rule-of-three upper bound, not a bootstrap interval, because a bootstrap of an all-zero count is degenerate.

A matched negative control. 1,999 randomly drawn gene pairs score a median e_{prox} of 0.026, with 0.55% above threshold—indistinguishable from the 0.42% base rate, as required.

The calibration

Figure 1 and Table 1 give the result. Redundancy is genuine evidence: the probability of interventional equivalence rises monotonically across the populated range, reaching 39× the base rate. It is also nowhere near identification: the ceiling is about one in six, and the trend does not continue into the extreme. Among the 13 most redundant pairs in the entire dataset, none are equivalent—a count small enough that it excludes only rates above roughly one in four, but one that gives no support to the idea that the curve keeps climbing.

For a practitioner the operational statement is: expression redundancy should raise suspicion that two genes are one hypothesis and should never settle it. At the redundancy levels where such judgements are typically made, four times in five or worse, the gene not chosen behaves differently.

Quantity	Estimate	95% CI
Base rate, all 43.9M pairs	0.42%	0.37–0.47
Ceiling, $r^2 \in [0.60, 0.70)$ (911 pairs)	17.0%	9.7–27.3
Lift over base rate	39.3×	23.0–64.8
Extreme redundancy, $r^2 \geq 0.8$	0 of 13	$\leq$ 23%

Table 1: **Headline calibration.** Intervals are gene-level bootstrap, $B = 100$; the last row is a rule-of-three bound.

Tissue identity manufactures redundancy

Correcting for cell lineage removes 56% of pairs at $r^2 \geq 0.6$ (2,317 → 1,023), and the fraction removed grows with redundancy: 37% at $r^2 \geq 0.4$, 42% at ≥ 0.5, about 70% at ≥ 0.7. Two genes that appear to track each other frequently track the tissue instead. A redundancy screen run on a multi-lineage panel without this correction is measuring cell-of-origin in substantial part, and increasingly so at exactly the redundancy levels that look most convincing.

Panel redundancy adds nothing over best-pair

The calibration asks about one partner. A natural objection is that real redundancy is many-to-one: a gene may be predictable from a *panel* of others without any single partner standing out. We tested it. For each gene we computed best-pair r_{obs}, panel R^2 from a ridge fit on its ten most correlated partners, and whether any partner is interventionally equivalent. The base rate of a gene having some equivalent partner is 0.542. Panel redundancy predicts that outcome with AUC 0.614; best-pair redundancy with AUC 0.607. At a matched cut of 0.6, best-pair gives $P = 0.735$ ($n = 623$) and panel gives $P = 0.677$ ($n = 2{,}687$). The pre-registered question is answered in the negative: the more elaborate statistic carries essentially the same information, and the simpler one is no worse.

Gene families show the opposite sign, as buffering predicts

Our pre-registered paralog flag was a proxy—a shared alphabetic symbol root. Replacing it with HGNC gene groups shows the proxy agrees only 70.3% of the time (3,644 false positives, 1,626 missed), which is worth knowing for any study that uses such a proxy. The published direction is unchanged: excluding real families, the ceiling is 26.3%, against 23.1% under the proxy.

The new observation is the contrast between family and non-family genes. Genes *inside* a gene family have a lower equivalence ceiling (13.9%, 388 pairs) than genes outside one (26.3%, 167 pairs). This is the sign that buffering predicts: paralogs that cover for one another are, under *single* knockout, measured as non-equivalent precisely because the compensation hides the phenotype. It is consistent with **(author?)** [1, 2] and we report it as convergence with their result rather than as an independent finding. It also marks the blind spot of this study honestly: for buffering pairs, our measure returns non-equivalence for a reason that has nothing to do with the genes being functionally different.

A subgroup analysis that failed, as declared

The pre-registration included a triple-negative breast cancer arm and declared in advance that it would be underpowered. It was, and we report it as such rather than rescuing it.

Two things went wrong before any result was produced, both instructive. The first receptor-low definition—lower tertile of ESR1, PGR and ERBB2 among breast lines—selected zero lines, because PGR's lower tertile *is* zero and a strict inequality against a threshold on the distribution's floor excludes exactly the receptor-negative lines it is meant to select. With an inclusive bound it selected 7. We then gated the definition itself against documented cell-line receptor status, using no dependency data: it recovered 4 of 19 known triple-negative lines. The definition failed its positive control, so the definition was replaced, not the biology reinterpreted. The replacement—ESR1 and PGR at or below the conventional not-expressed line, ERBB2 at or below the breast-panel median—recovers 10 of 19 and admits none of the 8 documented receptor-positive lines.

On the resulting 18 lines the calibration degenerates. 11.3% of pairs land above $r^2 = 0.6$, against 0.3% in the 50-line breast panel: with 18 samples r^2 is a noisy statistic and the observational axis inflates wholesale. Lift collapses from 11× (breast panel) to 3.3×. We draw no subgroup conclusion. The generalisable point is that a subgroup calibration is not a small version of a pan-cohort one; below a few dozen samples the redundancy axis stops carrying the information the curve requires.

Discussion

Expression redundancy is evidence about interventional equivalence, worth roughly a 39-fold change in odds, and it is not identification: at the redundancy levels where the judgement is usually made, the probability that two interchangeable-looking genes are actually interchangeable is about one in six at best. The practical consequence is not that redundancy should be ignored—a 39-fold lift is substantial—but that it should enter a shortlisting decision as a prior, with its magnitude known, rather than as a licence to collapse two hypotheses into one.

Three subsidiary findings sharpen that advice. Lineage correction is not optional: without it, a screen on a multi-lineage panel partly measures cell-of-origin, and the contamination is worst exactly where the redundancy looks most convincing. Panel redundancy buys nothing over the simplest pairwise statistic, so the additional modelling is not where the returns are. And gene-family membership moves the answer in the direction compensation predicts, which both corroborates the paralog-buffering literature and identifies where our own measure is structurally blind.

Limitations. Cell lines are not tumours, and knockout growth effect in culture is not clinical response. Nothing here nominates a target or supports a clinical claim. Single-gene knockouts are structurally blind to buffering, and the gene-family result quantifies rather than removes that blindness. The numerical ceiling depends on the equivalence threshold; the shape of the curve does not. The extreme-redundancy result rests on 13 pairs and should be read as an absence of support for continued climbing, not as a demonstration of zero. Chronos scores carry their own preprocessing assumptions, which we inherit.

Reuse. The calibration is cheap and portable. The whole analysis runs in about a minute on a consumer GPU against a public dataset, so it can be recomputed for any subtype, panel or release. More generally, the design transfers to any setting with paired observational and interventional measurements on the same system—methylation against knockout, proteomics against drug response, co-accessibility against CRISPRi—wherever a similarity-implies-equivalence step is currently assumed rather than measured.

Methods

Data. DepMap 24Q4: expression (`OmicsExpressionProteinCodingGenesTPMLogp1`) and CRISPR gene effect (Chronos, 4) for the 1,103 cell lines present in both, restricted to the 17,716 genes present in both matrices. All files are public and downloaded by the analysis script; none are redistributed.

Gene universe. Genes are retained if mean expression clears a floor and expression variance clears the declared percentile of expressed genes, leaving 9,368 genes and 43,875,028 unordered pairs. Filters were fixed in the pre-registration before any curve was computed.

Observational axis. r_{obs} is the squared Pearson correlation of expression across cell lines, after regressing out one-hot cell lineage; the uncorrected arm is reported alongside.

Interventional axis. The reported measure is proximity on *raw, uncentred* Chronos vectors,

$$e_{\text{prox}}(a, b) \;=\; \frac{2\langle k_a, k_b\rangle}{\langle k_a, k_a\rangle + \langle k_b, k_b\rangle},$$

which equals 1 for identical profiles and stays near 0 for unrelated ones. Centring, which Pearson performs, removes exactly the shared mean in which uniform co-essentiality lives; proximity keeps it. A pair counts as interventionally equivalent when $e_{\text{prox}} > 0.8$. Both axes are reported throughout.

Uncertainty. Gene pairs share genes, so pairs are not independent. All intervals are gene-level bootstrap, $B = 100$, resampling genes and recomputing the entire curve. Histograms are accumulated in double precision: the lowest-redundancy bin holds 36 million pairs, beyond the range over which single precision counts integers exactly.

Computation. Pair scoring is chunked on GPU so that no $G \times G$ matrix is materialised. The complete calibration runs in about one minute; the bootstrap adds about ten.

Supporting information

S1 Protocol. Pre-registration, fixing filters, measures, thresholds, controls and success criteria before any curve was computed.

S2 Ledger. The complete experiment ledger, including every voided result and both control failures described above.

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Competing interests

The author has declared that no competing interests exist.

Data availability

All data are public DepMap releases (https://depmap.org), downloaded by the analysis scripts and not redistributed. All analysis code, the pre-registration and the complete experiment ledger are archived at https://doi.org/10.5281/zenodo.21988145.

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